

ESP32-S3 Consolidated Pin Overview

Table 7-1. Consolidated Pin Overview

Pin No.	Pin Name	Pin Type	Pin Providing Power	Pin Settings		RTC IO MUX Function		Analog Function		IO MUX Function									
				At Reset	After Reset	F0	F3	F0	F1	F0	Type	F1	Type	F2	Type	F3	Type	F4	Type
1	LNA_IN	Analog																	
2	VDD3P3	Power																	
3	VDD3P3	Power																	
4	CHIP_PU	Analog	VDD3P3_RTC																
5	GPIO0	IO	VDD3P3_RTC	WPU, IE	WPU, IE	RTC_GPIO0	sar_i2c_scl_0			GPIO0	I/O/T	GPIO0	I/O/T						
6	GPIO1	IO	VDD3P3_RTC	IE	IE	RTC_GPIO1	sar_i2c_sda_0	TOUCH1	ADC1_CH0	GPIO1	I/O/T	GPIO1	I/O/T						
7	GPIO2	IO	VDD3P3_RTC	IE	IE	RTC_GPIO2	sar_i2c_scl_1	TOUCH2	ADC1_CH1	GPIO2	I/O/T	GPIO2	I/O/T						
8	GPIO3	IO	VDD3P3_RTC	IE	IE	RTC_GPIO3	sar_i2c_sda_1	TOUCH3	ADC1_CH2	GPIO3	I/O/T	GPIO3	I/O/T						
9	GPIO4	IO	VDD3P3_RTC			RTC_GPIO4		TOUCH4	ADC1_CH3	GPIO4	I/O/T	GPIO4	I/O/T						
10	GPIO5	IO	VDD3P3_RTC			RTC_GPIO5		TOUCH5	ADC1_CH4	GPIO5	I/O/T	GPIO5	I/O/T						
11	GPIO6	IO	VDD3P3_RTC			RTC_GPIO6		TOUCH6	ADC1_CH5	GPIO6	I/O/T	GPIO6	I/O/T						
12	GPIO7	IO	VDD3P3_RTC			RTC_GPIO7		TOUCH7	ADC1_CH6	GPIO7	I/O/T	GPIO7	I/O/T						
13	GPIO8	IO	VDD3P3_RTC			RTC_GPIO8		TOUCH8	ADC1_CH7	GPIO8	I/O/T	GPIO8	I/O/T			SUBSPICS1	O/T		
14	GPIO9	IO	VDD3P3_RTC		IE	RTC_GPIO9		TOUCH9	ADC1_CH8	GPIO9	I/O/T	GPIO9	I/O/T			SUBSPIHD	I/O/T	FSPiHD	I/O/T
15	GPIO10	IO	VDD3P3_RTC		IE	RTC_GPIO10		TOUCH10	ADC1_CH9	GPIO10	I/O/T	GPIO10	I/O/T	FSPII04	I/O/T	SUBSPICSO	O/T	FSPICSO	I/O/T
16	GPIO11	IO	VDD3P3_RTC		IE	RTC_GPIO11		TOUCH11	ADC2_CH0	GPIO11	I/O/T	GPIO11	I/O/T	FSPII05	I/O/T	SUBSPID	I/O/T	FSPID	I/O/T
17	GPIO12	IO	VDD3P3_RTC		IE	RTC_GPIO12		TOUCH12	ADC2_CH1	GPIO12	I/O/T	GPIO12	I/O/T	FSPII06	I/O/T	SUBSPICLK	O/T	FSPICLK	I/O/T
18	GPIO13	IO	VDD3P3_RTC		IE	RTC_GPIO13		TOUCH13	ADC2_CH2	GPIO13	I/O/T	GPIO13	I/O/T	FSPII07	I/O/T	SUBSPIQ	I/O/T	FSPiQ	I/O/T
19	GPIO14	IO	VDD3P3_RTC		IE	RTC_GPIO14		TOUCH14	ADC2_CH3	GPIO14	I/O/T	GPIO14	I/O/T	FSPIDQS	O/T	SUBSPIWP	I/O/T	FSPiWP	I/O/T
20	VDD3P3_RTC	Power																	
21	XTAL_32K_P	IO	VDD3P3_RTC			RTC_GPIO15		XTAL_32K_P	ADC2_CH4	GPIO15	I/O/T	GPIO15	I/O/T	UORTS	O				
22	XTAL_32K_N	IO	VDD3P3_RTC			RTC_GPIO16		XTAL_32K_N	ADC2_CH5	GPIO16	I/O/T	GPIO16	I/O/T	UOCTS	I1				
23	GPIO17	IO	VDD3P3_RTC		IE	RTC_GPIO17			ADC2_CH6	GPIO17	I/O/T	GPIO17	I/O/T	UITXD	O				
24	GPIO18	IO	VDD3P3_RTC		IE	RTC_GPIO18			ADC2_CH7	GPIO18	I/O/T	GPIO18	I/O/T	UIRXD	I1	CLK_OUT3	O		
25	GPIO19	IO	VDD3P3_RTC			RTC_GPIO19		USB_D-	ADC2_CH8	GPIO19	I/O/T	GPIO19	I/O/T	UIRTS	O	CLK_OUT2	O		
26	GPIO20	IO	VDD3P3_RTC	USB_PU	USB_PU	RTC_GPIO20		USB_D+	ADC2_CH9	GPIO20	I/O/T	GPIO20	I/O/T	UICTS	I1	CLK_OUT1	O		
27	GPIO21	IO	VDD3P3_RTC			RTC_GPIO21				GPIO21	I/O/T	GPIO21	I/O/T						
28	SPICST	IO	VDD_SPI	WPU, IE	WPU, IE					SPICST	O/T	GPIO26	I/O/T						
29	VDD_SPI	Power																	
30	SPIHD	IO	VDD_SPI	WPU, IE	WPU, IE					SPIHD	I/O/T	GPIO27	I/O/T						
31	SPIWP	IO	VDD_SPI	WPU, IE	WPU, IE					SPIWP	I/O/T	GPIO28	I/O/T						
32	SPICSO	IO	VDD_SPI	WPU, IE	WPU, IE					SPICSO	O/T	GPIO29	I/O/T						
33	SPICLK	IO	VDD_SPI	WPU, IE	WPU, IE					SPICLK	O/T	GPIO30	I/O/T						
34	SPIQ	IO	VDD_SPI	WPU, IE	WPU, IE					SPIQ	I/O/T	GPIO31	I/O/T						
35	SPID	IO	VDD_SPI	WPU, IE	WPU, IE					SPID	I/O/T	GPIO32	I/O/T						
36	SPICLK_N	IO	VDD_SPI/VDD3P3_CPU	IE	IE					SPICLK_P_DIFF	O/T	GPIO48	I/O/T	SUBSPICLK_P_DIFF	O/T				
37	SPICLK_P	IO	VDD_SPI/VDD3P3_CPU	IE	IE					SPICLK_N_DIFF	O/T	GPIO47	I/O/T	SUBSPICLK_N_DIFF	O/T				
38	GPIO33	IO	VDD_SPI/VDD3P3_CPU		IE					GPIO33	I/O/T	GPIO33	I/O/T	FSPiHD	I/O/T	SUBSPiHD	I/O/T	SPII04	I/O/T
39	GPIO34	IO	VDD_SPI/VDD3P3_CPU		IE					GPIO34	I/O/T	GPIO34	I/O/T	FSPICSO	I/O/T	SUBSPICSO	O/T	SPII05	I/O/T
40	GPIO35	IO	VDD_SPI/VDD3P3_CPU		IE					GPIO35	I/O/T	GPIO35	I/O/T	FSPID	I/O/T	SUBSPID	I/O/T	SPII06	I/O/T
41	GPIO36	IO	VDD_SPI/VDD3P3_CPU		IE					GPIO36	I/O/T	GPIO36	I/O/T	FSPICLK	I/O/T	SUBSPICLK	O/T	SPII07	I/O/T
42	GPIO37	IO	VDD_SPI/VDD3P3_CPU		IE					GPIO37	I/O/T	GPIO37	I/O/T	FSPiQ	I/O/T	SUBSPiQ	I/O/T	SPIDQS	I/O/T
43	GPIO38	IO	VDD3P3_CPU		IE					GPIO38	I/O/T	GPIO38	I/O/T	FSPiWP	I/O/T	SUBSPIWP	I/O/T		
44	MTCK	IO	VDD3P3_CPU		IE					MTCK	I1	GPIO39	I/O/T	CLK_OUT3	O	SUBSPICS1	O/T		
45	MTDO	IO	VDD3P3_CPU		IE					MTDO	O/T	GPIO40	I/O/T	CLK_OUT2	O				
46	VDD3P3_CPU	Power																	
47	MTDI	IO	VDD3P3_CPU		IE					MTDI	I1	GPIO41	I/O/T	CLK_OUT1	O				
48	MTMS	IO	VDD3P3_CPU		IE					MTMS	I1	GPIO42	I/O/T						
49	UOTXD	IO	VDD3P3_CPU	WPU, IE	WPU, IE					UOTXD	O	GPIO43	I/O/T	CLK_OUT1	O				
50	UORXD	IO	VDD3P3_CPU	WPU, IE	WPU, IE					UORXD	I1	GPIO44	I/O/T	CLK_OUT2	O				
51	GPIO45	IO	VDD3P3_CPU	WPD, IE	WPD, IE					GPIO45	I/O/T	GPIO45	I/O/T						
52	GPIO46	IO	VDD3P3_CPU	WPD, IE	WPD, IE					GPIO46	I/O/T	GPIO46	I/O/T						
53	XTAL_N	Analog																	
54	XTAL_P	Analog																	
55	VDDA	Power																	
56	VDDA	Power																	
57	GND	Power																	

* For details, see Section 2 Pins. Regarding highlighted cells, see Section 2.3.4 Restrictions for GPIOs and RTC_GPIOs.